

G-1

1. **Explain the key objectives of 5G mobile communication systems. (3 marks)**
 - Provide very high data rates (up to tens of Gbps).
 - Ensure ultra-low latency for real-time applications.
 - Support massive device connectivity (IoT).
 - Improve reliability, flexibility, and user experience.
2. **Describe the role of New Radio (NR) in 5G architecture. (3 marks)**
 - NR is the new air interface standard for 5G.
 - It enables high-speed, low-latency communication.
 - Supports multiple frequency bands including sub-6 GHz and mmWave.
 - Facilitates advanced technologies like massive MIMO.
3. **What are the three main usage scenarios defined for 5G? Briefly explain each. (3 marks)**
 - **eMBB (enhanced Mobile Broadband):** High data rates for applications like video streaming.
 - **URLLC (Ultra-Reliable Low Latency Communications):** Critical applications needing low delay (e.g., surgery, autonomous vehicles).
 - **mMTC (massive Machine-Type Communications):** Supports large numbers of IoT devices.
4. **Discuss the importance of network slicing in 5G networks. (3 marks)**
 - Allows creation of multiple virtual networks on a single physical infrastructure.
 - Each slice is optimized for specific applications or services.
 - Improves efficiency, flexibility, and resource utilization.
5. **What are the main components of the 5G system architecture? (3 marks)**
 - User Equipment (UE)
 - Next Generation Radio Access Network (NG-RAN)
 - 5G Core Network (5GC)
 - Application Servers (AS) These components work together to deliver communication services.
6. **Define peak data rate (PDR) and explain its significance in 5G performance. (3 marks)**
 - PDR is the maximum achievable data transmission rate under ideal conditions.
 - It depends on bandwidth and spectral efficiency.
 - Indicates the maximum capacity and speed of the network.
7. **What is user plane latency and why is it important in 5G applications? (3 marks)**
 - It is the time taken for data to travel from sender to receiver.
 - Critical for real-time services like gaming, VR, and remote surgery.
 - Lower latency improves responsiveness and user experience.
8. **Explain the concept of connection density in 5G networks. (3 marks)**
 - Refers to the number of devices that can be connected per unit area.
 - 5G supports up to 1 million devices per km².
 - Essential for IoT and smart city applications.
9. **Describe the differences between Standalone (SA) and Non-Standalone (NSA) architectures in 5G. (3 marks)**
 - NSA: Uses existing 4G infrastructure with 5G NR; faster deployment.
 - SA: Fully independent 5G system with its own core network.
 - SA provides better performance and full 5G capabilities.
10. **Why is spectrum allocation important for 5G deployment? (3 marks)**
 - Determines network capacity, speed, and coverage.
 - Different bands (low, mid, high) serve different purposes.
 - Efficient allocation ensures optimal performance and supports diverse ap.

11. Why is the transition from 5G to 6G networks considered essential for the future?

The transition to 6G is driven by the limitations of 5G in handling the rapid growth of data traffic and the rise of “Internet of Everything” (IoE) applications. Although 5G is already in use, it cannot support the expected increase in global mobile data traffic, which may exceed 5,000 Exabytes per month by 2030.

Future IoE networks will include autonomous vehicles, advanced healthcare, and delay-sensitive intelligent services. These require continuous sensing and computing capabilities beyond what 5G can provide.

Therefore, 6G is being developed to support extreme data rates, ultra-high reliability, and intelligent services that previous generations cannot achieve.

12. Detail the primary performance visions and key indicators expected for 6G networks.

The 6G vision introduces an “ultra era” with major improvements over 5G performance indicators (KPIs). While 5G supports peak data rates of 20 Gbps, 6G aims to exceed 100 Gbps.

Other important visions include:

- **Latency and Reliability:** End-to-end delay of less than 1 ms with ultra-high reliability for critical applications.
- **Energy and Mobility:** 10 times better energy efficiency than 5G and support for mobility speeds up to 1000 km/h.
- **Universal Coverage:** Global connectivity by combining terrestrial networks with non-terrestrial platforms like satellites and UAVs.

13. Explain the concept and significance of “Smart Radio Environments” in 6G.

6G shifts focus from massive **MIMO** (used in 5G) to “reconfigurable intelligent reflecting surfaces” (RIS).

Smart radio environments involve covering physical objects with meta-materials that can control and shape radio wave propagation. This allows the network to actively adjust the environment to:

- Improve connectivity
- Increase data rates
- Reduce energy consumption

By making the wireless environment intelligent and controllable, 6G can overcome signal obstacles and provide the high reliability needed for future services.

14. How does 6G intend to achieve global and seamless network integration?

6G aims to seamlessly integrate **terrestrial** and **non-terrestrial** networks, including airborne, underwater, and space-based systems.

This includes:

- **Satellite and Aerial Platforms:** Use of Low Earth Orbit (LEO) satellites, high-altitude platforms (HAPs), and unmanned aerial vehicles (UAVs) to ensure coverage in remote areas.
- **Undersea and Maritime:** Extending connectivity to oceans and underwater environments.
- **Enabling Frameworks:** Technologies like network slicing, multi-access edge computing (MEC), and software-defined networking (SDN) ensure a unified and efficient global network.

15. Discuss the roles of Artificial Intelligence (AI) and Machine Learning (ML) as key enablers for 6G.

AI and ML are essential for making 6G networks intelligent and automated. Their roles include:

- **Big Data Management:** Analyzing and compressing large volumes of data from real-time sensing devices.
- **Network Optimization:** Enabling self-learning, self-configuration, and dynamic resource allocation.
- **Complexity Reduction:** Replacing traditional algorithms to reduce complexity and improve performance.
- **Intelligent Edge:** Supporting predictive analytics and fast responses at both network core and edge.

16. What is Rate Splitting Multiple Access (RSMA), and how does it improve upon existing techniques?

RSMA is a **non-orthogonal** multiple access technology designed for 6G. It acts as a bridge between Space Division Multiple Access (SDMA) and Non-Orthogonal Multiple Access (NOMA).

Key advantages:

- **Flexible Interference Management:** Treats part of interference as noise (like SDMA) and decodes another part (like NOMA).
- **Technical Implementation:** Uses precoding at the transmitter and Successive Interference Cancellation (SIC) at receivers.
- **Efficiency:** Improves signal decoding and interference handling, making it suitable for complex multi-user environments.

17. How will 6G address the advanced security, secrecy, and privacy requirements of future users?

6G focuses strongly on security and privacy due to increased data usage and network integration. Key technologies include:

- **Quantum Communication:** Provides highly secure communication using quantum principles, resistant to eavesdropping.
- **Blockchain Technology:** Ensures trust, secrecy, and privacy through decentralized and immutable data systems.
- **AI-Driven Security:** AI and ML help detect and prevent threats in real-time within the 6G network.

G-3

18. Why is a new RAN architecture needed for 5G?

A new RAN architecture is needed because 5G must support much higher traffic, more connected devices, and better service quality than older networks. Legacy **RAN** systems cannot efficiently handle all these demands.

- 5G needs high data rate and very low latency.
- It must support massive user density and many new applications.
- Operators also want better spectrum efficiency and lower CAPEX/OPEX.

So, redesigning the RAN is necessary to meet future mobile and industrial requirements.

19. What are the main RAN architectures discussed in the paper?

The paper mainly discusses four important RAN architectures for 5G. These are proposed as modern solutions to improve network performance and efficiency.

- Cloud-RAN (C-RAN)
- Heterogeneous Cloud-RAN (H-CRAN)
- Virtualized Cloud-RAN (V-CRAN)
- Fog-RAN (F-RAN)

These architectures are compared in terms of cost, energy, resource allocation, and system performance.

20. What are the three main service categories of 5G?

The **International Telecommunication Union (ITU)** divides 5G services into three major categories based on different communication needs.

- **eMBB** for high-speed broadband services
- **mMTC** for massive machine and device connections
- **URLLC** for highly reliable and very low-latency applications

These three categories together allow 5G to support both normal users and industrial use-cases.

21. What is eMBB in 5G?

eMBB means **enhanced Mobile Broadband**. It is designed to provide faster and better internet service for ordinary users.

- It supports high-bandwidth applications.
- It is useful for web browsing, HD video streaming, and virtual reality.
- It improves user experience by giving higher speed and better connectivity.

So, eMBB mainly focuses on strong broadband performance in 5G networks.

22. What are the key enabling technologies of 5G?

5G depends on several important technologies from both networking and wireless communication fields. These technologies help improve performance, flexibility, and service delivery.

- In networking: MEC, SDN, NFV, and network slicing
- In wireless: **massive MIMO**, ultra-dense networking, mmWave, and D2D communication
- These technologies help increase capacity, reduce delay, and support many types of applications

So, key enablers make advanced 5G operation possible.

23. What is C-RAN in 5G networks?

C-RAN stands for Cloud Radio Access Network. In this architecture, baseband processing units are separated from remote radio heads and pooled in a centralized cloud-like BBU pool.

- RRHs stay near users, while BBUs are centralized
- This makes resource sharing easier
- It improves scalability and simplifies management
- It also supports efficient network operation for 5G

So, C-RAN is a more centralized and flexible RAN design for modern mobile networks.

24. What are the benefits of C-RAN in 5G?

C-RAN provides several advantages that make it attractive for 5G deployment. It improves both technical performance and operational efficiency.

- It reduces CAPEX and OPEX
- It lowers energy consumption
- It improves spectral efficiency and network throughput
- It simplifies network management and maintenance
- It also helps in load balancing and network scalability

Because of these benefits, C-RAN is considered an important architecture for 5G systems.

G-4

25. What is millimeter-wave (mmWave) technology and why is it important in 5G/6G?

Millimeter-wave (mmWave) technology refers to the use of electromagnetic waves with frequencies typically ranging from 30 GHz to 300 GHz. These waves have very short wavelengths (1–10 mm), which allows the use of compact antenna arrays.

In 5G and future 6G systems, mmWave is important because it provides **very large available spectrum** compared to traditional sub-6 GHz bands. This enables **ultra-high data rates (Gbps level), low latency, and improved network capacity**.

Moreover, mmWave allows advanced technologies like **massive MIMO and beamforming**, which improve spectral efficiency and support a large number of users simultaneously. Therefore, mmWave is a key enabler for next-generation wireless communication systems.

26. What are the three main application scenarios of 5G?

The International Telecommunication Union (ITU) has defined three main application scenarios for 5G:

1. **Enhanced Mobile Broadband (eMBB):**
This focuses on providing **very high data rates** for applications such as HD video streaming, virtual reality (VR), and augmented reality (AR).
2. **Massive Machine-Type Communication (mMTC):**
This supports **a large number of connected devices**, such as IoT sensors, smart homes, and smart cities. It emphasizes scalability and energy efficiency.
3. **Ultra-Reliable Low Latency Communication (URLLC):**
This is designed for **mission-critical applications** requiring extremely low delay and high reliability, such as autonomous vehicles, remote surgery, and industrial automation.

These three scenarios together make 5G versatile and capable of supporting diverse applications.

27. What are the main differences between 4G and 5G?

There are several key differences between 4G and 5G:

- **Data Rate:** 4G provides speeds in the range of Mbps, whereas 5G can achieve **Gbps-level speeds**.
- **Latency:** 4G latency is higher (tens of milliseconds), while 5G reduces it to **around 1 ms**, enabling real-time communication.
- **Frequency Bands:** 4G mainly operates below 6 GHz, while 5G also uses **mmWave frequencies (24–52 GHz)**.
- **Capacity:** 5G supports a much larger number of connected devices due to improved spectral efficiency.
- **Technologies Used:** 5G uses advanced technologies like **massive MIMO, beamforming, ultra-dense networks**, and full-duplex communication.

Overall, 5G provides faster, more reliable, and more efficient communication compared to 4G.

28. What are the two main reasons for using mmWave frequencies in 5G?

The use of mmWave frequencies in 5G is mainly due to two reasons:

- **Spectrum Scarcity in Lower Bands:**

The sub-6 GHz frequency bands are already heavily used by existing communication systems such as cellular networks, Wi-Fi, and satellite communication. Therefore, it is difficult to find additional bandwidth in these bands.

- **Large Available Bandwidth in mmWave:**

mmWave frequencies offer **much wider bandwidth**, which allows higher data rates and better performance. Even with the same relative bandwidth, the absolute bandwidth at mmWave is much larger than at lower frequencies.

These factors make mmWave an ideal choice for meeting the high data demands of 5G networks.

29. What is beamforming and why is it needed in mmWave communication?

Beamforming is a signal processing technique in which multiple antennas are used to direct the transmission or reception of signals in a specific direction instead of broadcasting them in all directions.

In mmWave communication, signals experience high path loss and are easily blocked by obstacles. Beamforming helps to overcome these issues by focusing energy toward the intended user, thereby improving signal strength and coverage.

Additionally, beamforming increases signal-to-noise ratio (SNR), enhances data rates, and reduces interference. It is therefore an essential technique for efficient mmWave communication in 5G and 6G systems.

30. What are the three main beamforming architectures?

Beamforming architectures can be classified into three main types:

- **Analog Beamforming:**

In this method, phase shifting is done in the analog domain using phase shifters. It is **low-cost and simple**, but supports only one beam at a time and has limited flexibility.

- **Digital Beamforming:**

Each antenna element is connected to a separate RF chain and processed digitally. This allows **high flexibility, multiple beams, and better performance**, but it is expensive and consumes more power.

- **Hybrid Beamforming:**

This combines both analog and digital beamforming. It reduces the number of RF chains while still providing good performance. It offers a **balance between cost, complexity, and efficiency**, and is widely used in practical 5G systems.

31. What is the main goal of 6G and how is it different from 5G?

The main goal of 6G is to develop an **intelligent, global communication network** that extends beyond terrestrial networks.

Unlike 5G, which mainly covers land-based communication, 6G will provide **3D connectivity**, integrating **satellites, aircraft, drones, and ocean networks**. It aims to cover the entire Earth and near outer space.

Additionally, 6G is expected to:

- Provide even higher data rates than 5G
- Use advanced technologies like **terahertz communication and artificial intelligence (AI)**
- Support new applications such as holographic communication and fully immersive virtual environments

Thus, 6G represents a significant evolution from 5G in terms of coverage, intelligence, and performance.

G-5

32. What is the main goal of 6G technology? (2 marks)

- To create a fully connected cyber-physical world combining communication, computing, and intelligence.
- To provide advanced services like real-time sensing, AI support, and immersive experiences.

33. Why will 5G be insufficient in the future? (2 marks)

- 5G cannot meet the future demand for higher speed and ultra-low latency.

- New technologies like automation and immersive communication will require 6G capabilities.

34. What are the main drivers for the development of 6G? (3 marks)

- Trustworthy and secure communication systems.
- Sustainability and efficient use of resources.
- AI-based automation and global connectivity.

35. Explain the concept of a cyber-physical world. (3 marks)

- A system where physical objects and digital systems are closely connected.
- Real-world data is continuously collected and used in digital platforms.
- It enables real-time monitoring, control, and interaction.

36. How will AI influence future networks? (3 marks)

- AI will automate network management and improve efficiency.
- It will analyze large data and optimize performance.
- It will support smart decision-making with less human effort.

37. What are the key capabilities expected from 6G networks? (4 marks)

- Extremely high data speed (hundreds of Gbps) and ultra-low latency.
- Massive device connectivity with high reliability.
- Global coverage including advanced sensing and positioning features.
- Strong AI integration for smart network control.

38. Describe the importance of sustainability in 6G. (4 marks)

- It reduces energy consumption and environmental impact.
- It ensures efficient use of network resources.
- It supports smart systems like agriculture and energy management.
- It helps achieve global sustainability goals.

39. What is meant by global 3D connectivity in 6G? (4 marks)

- It provides network access everywhere including land, air, and sea.
- It uses satellites, drones, and high-altitude platforms.
- It ensures connectivity in remote and rural areas.
- It helps reduce the digital divide.

40. Explain the role of cognitive networks in 6G. (5 marks) intelligent network

- Cognitive networks use AI to manage and optimize networks automatically.
- They learn from data and adapt to changing conditions.
- They improve performance and reduce operational costs.
- They support autonomous and intelligent network systems.
- They provide better service quality with less human control.

41. How will 6G transform society in the future? (5 marks)

- It will enable smart cities, autonomous transport, and healthcare systems.
- It will connect physical and digital worlds in real time.
- It will improve productivity and daily life efficiency.
- It will increase automation and intelligent services globally.
- It will create a highly connected and advanced society

G-6

42. Compare the integration of IoT support in 4G and 5G with the proposed approach for 6G.

In 4G and 5G, IoT support was added later, not from the beginning. At first, both networks mainly focused on improving mobile internet speed.

- In 4G, the main goal was mobile broadband. Support for low-power IoT devices (like smart meters and tracking devices) was added much later through **LTE-M and NB-IoT**.
- In 5G, the first version also focused on faster internet. Later, in Release 17, support for simpler IoT devices (like wearables and sensors) was added through **RedCap**.

However, for 6G, the idea is different. The paper suggests that 6G should support all types of services — from small IoT devices to high-speed mobile internet — right from the first release, instead of adding them later.

43. What were the key drawbacks of the earlier generations, and how does the paper suggest 6G should address them?

Drawbacks with the 4G and 5G Approach

Adding IoT support later in 4G and 5G caused several problems:

- **More Complexity**
Since IoT was added later, the system became more complicated. Some technologies (like NB- IoT and LTE-M) worked almost like separate systems instead of being fully integrated. This made the network harder to design and manage.
- **Fragmented Standards**
New features were added step by step, which made the overall standard less unified. This created confusion and made network management more difficult.
- **Slow Market Growth**
Because IoT support was not available from the beginning, industries had to wait years for proper IoT solutions. This delayed large-scale adoption.
- **Higher Costs**
Since everything was not unified from the start, devices could not be produced at large scale quickly. This reduced the benefit of mass production and kept costs higher.

6G Proposed Approach:

- **Support all devices from the start** – 6G will support both simple IoT devices and high- speed internet devices from day one.
- **One unified system** – It will use one flexible network system instead of adding separate solutions later.
- **Clear low-cost design** – Simple device features will be defined early to reduce cost.
- **Flexible for new devices** – It will easily support mid-range and future smart devices.

44. According to the paper, what are the recommended low-end device capabilities for 6G, and why is 5 MHz considered a suitable bandwidth for such devices?

For simple IoT devices (like smart meters and trackers), the paper suggests:

- 5 MHz bandwidth
- 1 receive antenna
- Half-duplex FDD (can send or receive at one time, not both)

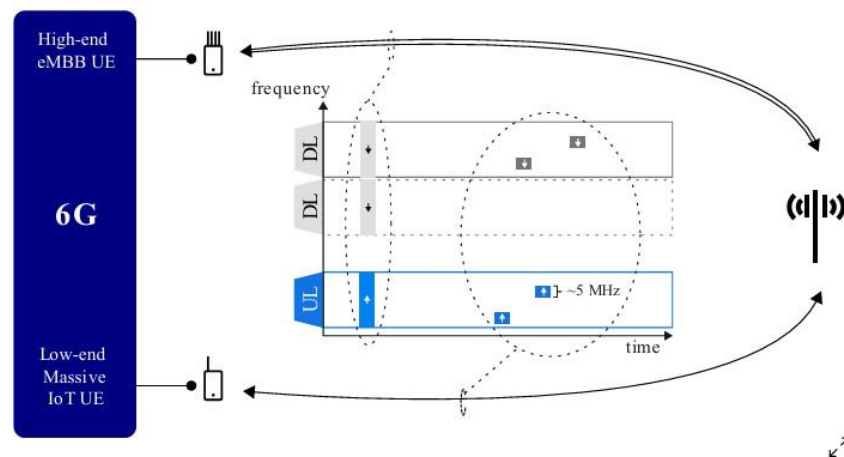


Fig: Difference between high-end and low-end 6G device capabilities (FDD).

Why 5 MHz is a Good Choice

- **Saves Cost (Enough)**
Reducing bandwidth below 5 MHz does not reduce cost much more. So going lower is not worth it.
- **Better Performance**
5 MHz gives:
 - ✓ Better coverage (works better in basements or remote areas)
 - ✓ Faster data speed (quick updates)
 - ✓ Longer battery life (finishes sending data faster)
- **Does Not Affect High-End Devices**

If bandwidth is too small, it could reduce performance for advanced smartphones. 5 MHz is wide enough to avoid this problem.

- Matches Network Spectrum

Most networks already use bandwidths larger than 3 MHz, so 5 MHz fits well.

45. Explain the concept of “half-duplex FDD” and how it contributes to reducing device cost in 6G. Why is this important for massive IoT deployments?

Half-Duplex FDD is a communication mode where a device can either transmit or receive at one time, but not both at the same time, even though it uses separate frequencies for sending and receiving. First, understand the difference:

Full-Duplex FDD

FDD = Frequency Division Duplex

- The device can send and receive at the same time.
- It uses different frequencies for sending and receiving.
- It needs a special and expensive duplex filter to prevent signal interference.
- More frequency bands = more filters = higher cost.

Half-Duplex FDD

- The device cannot send and receive at the same time.
- It either sends or receives, one at a time.
- It does not need the expensive duplex filter.
- A simple, low-cost switch is enough.

How It Reduces Device Cost

Half-duplex FDD removes the need for expensive duplex filters. Instead, it uses one simple and cheap switch. This makes the modem:

- Simpler
- Cheaper to manufacture
- Lower in total hardware cost (BoM)

Why This Is Important for Massive IoT

Massive IoT means deploying thousands or millions of devices, such as:

- Smart electricity meters
- Water leak sensors
- Asset tracking tags
- Smart streetlights

Since these devices are used in very large numbers, they must be very low-cost.

Even saving a small amount of money per device can save millions of dollars in large deployments.

So, half-duplex FDD helps make large-scale IoT affordable and practical.

46. What role do high-end device capabilities play in 6G, and how does the paper propose to balance support for both high-end and low-end devices in a single network?

Role of High-End 6G Devices

High-end 6G devices, like flagship smartphones, XR headsets, and fixed wireless access points, are built for extreme performance. They enable:

- **Immersive Communication:** Streaming 8K video to XR devices.
- **Time-Critical Communication:** Apps needing very low delay and high reliability.
- **High-Capacity Connectivity:** Multi-gigabit data speeds.

Key Features Needed:

- **Wide Bandwidth:** At least 200 MHz, including new cmWave bands (7–15 GHz).
- **Massive MIMO:** Thousands of antennas with advanced beamforming.
- **Carrier Aggregation:** Combine multiple frequency bands for faster speeds.
- **Uplink-Downlink Decoupling:** Send on a low band (good coverage) and receive on a high band (fast data).

Balancing High-End and Low-End Devices in 6G:

The paper suggests **three strategies** to make 6G work for both cheap IoT devices and expensive high-end devices in a single network:

- **Single, Scalable Network (RAT)**
 - ✓ One common network handles all devices.

- ✓ Low-end devices use a smaller subset of the features.
- ✓ High-end devices use the full capabilities.
- **Avoid Extremes That Hurt Performance**
 - ✓ Low-end devices get 5 MHz bandwidth instead of too narrow (like 1 MHz).
 - ✓ This prevents the network from slowing down high-end devices.
- **Support a Continuum of Devices**
 - ✓ 6G doesn't just support the cheapest and most expensive devices.
 - ✓ It also allows mid-range devices (like smart AR glasses) to work using standard features.
 - ✓ This speeds up innovation and market deployment.

47. Explain the concept of “device complexity vs. device bandwidth” as illustrated in Figure 4 of the paper. What conclusion does the paper draw from this relationship, and how does it influence the recommended low-end bandwidth for 6G?

In Figure 4 in the paper shows a graph that compares **device bandwidth** (how much frequency spectrum a device can use) with **device complexity** (which directly affects the cost of the device's modem). The main idea is to find out how much cost we can save by making the bandwidth smaller.

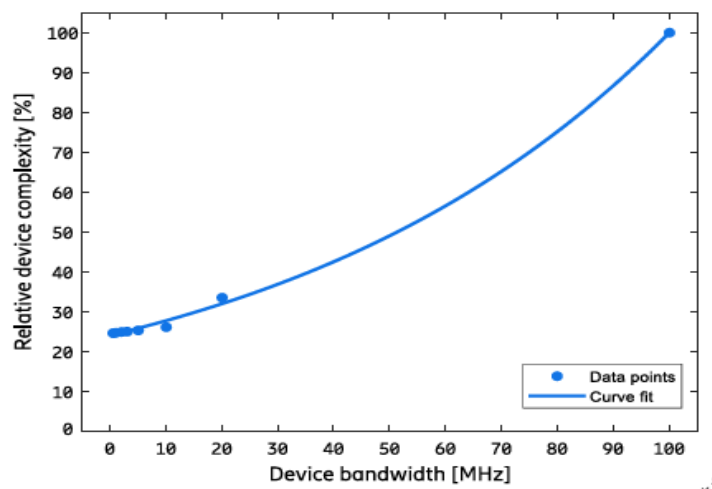


Fig 4: Relation between device complexity and device bandwidth

What did the graph reveal?

The graph shows an important trend:

- When you reduce bandwidth from a high value down to about 5 MHz, the device complexity (and cost) drops significantly.
- However, if you go below 5 MHz (for example, down to 1 MHz or 200 kHz), the cost reduction becomes very small—only a few percent.
- In other words, the curve flattens out below 5 MHz. You are making the device much less capable, but you are hardly saving any more money.

What conclusion does the paper draw?

The paper concludes that there is **no good reason** to support extremely narrow bandwidths (like 1 MHz or less) in 6G. The tiny cost saving is not worth the loss in performance. As the paper states, going below 5 MHz “would lower device complexity by barely a few percent.”

How does this influence the 6G recommendation?

Because of this finding, the paper **recommends 5 MHz** as the ideal low-end bandwidth for 6G IoT devices. It provides the best balance: you get the majority of the cost savings, but you also keep good enough performance for coverage, data rate, and battery life.

48. What is "Ambient IoT," and why does the paper exclude it from further discussion as a cellular solution for 6G?

Ambient IoT refers to a type of IoT device that is **extremely simple and low-power**. These devices often do **not have batteries** and instead **harvest energy from the environment** (like radio waves, light, or movement). They are used for very simple sensing and tracking applications.

The paper mentions Ambient IoT only for completeness but then clearly states it will not discuss it further as a cellular solution. The reason is simple:

- In 3GPP Release 19, the maximum distance allowed between an Ambient IoT device and a base station is only 15 meters.
- This is far too short for typical cellular IoT use cases, where devices need to connect over kilometers (e.g., a water meter in a basement or a tracker on a shipping container).
- Since 6G is meant to be a wide-area cellular network, a technology limited to 15 meters is not suitable.

Therefore, the paper focuses only on cellular IoT solutions that can work over long distances, leaving Ambient IoT for short-range technologies like RFID or local connectivity.

49. According to the paper, what are the key performance indicators (KPIs) that are crucial for massive IoT devices in 6G? Explain why each one matters for real-world IoT deployments.

For massive IoT devices (like smart meters, trackers, and sensors), data speed is not the main concern. Instead, the paper highlights several other KPIs that are much more important:

KPI	Why It Matters for Real-World IoT
Long Device Battery Life	IoT devices are often deployed in hard-to-reach places (e.g., underground water pipes, remote fields). Changing batteries frequently is impossible or too expensive; devices need to last 10–20 years on a single battery.
Extensive Coverage	IoT devices are not always in perfect locations. They may be in basements, behind walls, or deep inside buildings. The network must reach them reliably.
GPS-less Positioning	GPS consumes a lot of battery and doesn't work indoors. IoT devices need to determine their location using the cellular network itself (e.g., for asset tracking) without needing a dedicated GPS chip.
Efficient Transmission of Small Data	IoT devices usually send very small amounts of data (e.g., a temperature reading or meter value). The network must handle these tiny transmissions efficiently without wasting signaling overhead.
High Capacity	In a smart city, there could be thousands or even millions of devices in a small area. The network must support this massive number of connections simultaneously.
Longevity of Service Support	IoT deployments last for decades. The technology must be supported for a very long time so that devices don't become obsolete while still in the field.

All these KPIs ensure that massive IoT works reliably, affordably, and for many years without manual intervention.

50. How does the paper suggest 6G should handle "in-between" devices—those that are neither low-end massive IoT nor high-end eMBB? Provide examples of such devices and explain why this approach is beneficial.

"In-between" devices are those that need more capability than a simple sensor but do not require the full power of a flagship smartphone. They fall somewhere in the middle of the capability spectrum.

Examples of "In-Between" Devices:

- **AI-Driven Lightweight AR Glasses:** These need decent data rates and low latency for a smooth experience, but they don't need 8K video streaming capabilities.
- **Industrial Sensors with Camera:** A sensor in a factory that occasionally sends images along with data needs more bandwidth than a simple temperature sensor.
- **Wearable Health Monitors:** Devices that track heart rate, send alerts, and maybe do a video call with a doctor in an emergency.

The paper proposes that the 6G standard should enable all possible capabilities in between the low and high ends. This means:

- The standard will include a wide range of parameters and features.
- It will be left to the market to decide which specific combinations of capabilities are actually built and sold.
- There will be no gaps in support—any capability level between the two extremes is technically possible.

This approach beneficial for the following reasons:

1. **Faster Time-to-Market:** When a new type of device (like AR glasses) becomes popular, the required capabilities are already in the standard. Companies can start building them immediately without waiting for a new 3GPP release.

2. **Flexibility:** Manufacturers can choose the exact capability level that balances performance and cost for their specific product.
3. **Innovation:** It allows new and unexpected device types to emerge quickly because the standard does not restrict them to only two categories.

51. What role does "economy of scale" play in reducing IoT device costs, according to the paper?

How does this argument support the choice of a 5 MHz device bandwidth over a narrower 1 MHz bandwidth?

Economy of scale means that when you produce something in **very large quantities**, the cost per unit goes down. This happens because the cost of designing the product, setting up factories, and creating the supply chain is spread over millions of units.

How it applies to IoT devices:

If a particular type of device can be used for **many different use cases**, it will be produced and sold in huge volumes. This high volume pushes the price down further, making it even more attractive for more use cases—a positive cycle.

How this supports the 5 MHz choice:

The paper argues that a 5 MHz device can address **many different use cases**—from smart meters to wearables to basic industrial sensors. Because it is versatile, it will be produced in larger volumes, leading to lower costs through economy of scale.

On the other hand, a device with a very narrow 1 MHz bandwidth would only be suitable for a small subset of the simplest use cases. Its production volume would be lower, so it would not benefit as much from economy of scale. As the paper says, use cases that could work with 1 MHz would be "better to address with a somewhat higher device bandwidth" (like 5 MHz) because the small extra cost is worth the much larger market potential.

G-7

52. Why is 6G considered a "once-in-a-decade opportunity" for energy improvement?

6G is called a "once-in-a-decade opportunity" because a new mobile generation comes only after many years, so it gives a rare chance to make big improvements.

In older networks (like 4G and 5G), many important functions are already fixed, so it is hard to reduce energy use. But in 6G, these can be redesigned from the beginning.

- It can improve idle-mode functions (like paging and system signals), which use a lot of energy.
- It helps reduce wasted energy when there is no data traffic.
- The improvements will last for many years because network systems last for a long time.

So, 6G is a special chance to save a large amount of energy in networks for the future.

53. Explain the role of idle-mode in energy consumption.

Idle-mode means the mobile is not using internet or data, but it is still connected to the network. Even in this state, the network keeps sending basic signals like:

- Paging (for calls/messages)
- System information
- Random access signals

These signals are sent continuously, even when there is no data usage. Since most users are idle most of the time, the network uses a lot of energy **just to stay ready, not for real communication**. This causes continuous energy waste, especially in large networks. In fact, idle-mode energy cost is often higher than active data transmission. In fact, idle-mode energy cost is often higher than active data transmission. So reducing idle-mode signals can save a lot of energy.

54. What are traffic characteristics in cellular networks?

Traffic characteristics in cellular networks are-

- Most data sessions are small in size.
- Only 1% sessions are large, but they carry most of the data.
- Traffic is bursty, not continuous.
- Many cells have no users at a given time.
- This shows that networks are underutilized most of the time.

55. What is micro-sleep in 6G networks?

Micro-sleep is **when network components temporarily turn off** during idle periods. It happens between signal transmissions. Lean design helps increase these sleep periods.

- This reduces unnecessary energy usage.
- It is a key technique for saving power in low traffic conditions.

56. Explain the importance of separating idle-mode and connected-mode functions.

The importance of separating idle-mode and connected-mode functions are -

- Idle-mode and connected-mode have different requirements. Combining them increases signal complexity and energy usage. Separation allows optimized design for each function.
- Idle-mode can use low-energy signals.
- Connected-mode can focus on performance and speed.

57. What is the benefit of centralized RAN processing in 6G?

The benefits of centralized RAN processing in 6G are-

- Processing is moved to a central location instead of many sites.
- Enables resource sharing and better coordination.
- Reduces hardware duplication.
- Improves energy efficiency and flexibility.
- Makes the network more scalable and adaptive.

58. What is requirement management in 6G networks?

Requirement management in 6G networks means **managing network resources based on user needs** and service demands. It converts requirements like speed, latency, priority into hardware settings.

- Helps **control power usage** of radio units and processors.
- Prevents unnecessary high energy usage.
- Ensures energy is used only when and where needed.

59. What is 6G and what is its main goal?

6G is the sixth generation of mobile network technology, coming after 5G. It is expected to be introduced around 2030. The main goal of 6G is to provide extremely fast data speed, very low latency, and highly reliable communication. It will connect the digital and physical worlds more closely, enabling advanced applications like smart cities, remote work, virtual reality, and intelligent systems. It also focuses on better energy efficiency and sustainability.

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60. What is RAN and Core Network in 6G?

RAN (Radio Access Network) is the part of the network that connects user devices (like smartphones) to the main network using wireless signals. The Core Network (CN) is responsible for managing data flow, security, and services. In 6G, RAN will become more intelligent and flexible using AI, while the Core Network will be cloud-based and more efficient. Together, they ensure smooth communication and service delivery.

61. What is cloud-native design and why is it important in 6G?

Cloud-native design means building network services using cloud computing technologies instead of traditional hardware systems. In 6G, this allows network functions to run as software in containers. It is important because it provides flexibility, easy updates, faster deployment, and better scalability. It also helps in reducing costs and improving performance by using modern cloud infrastructure.

62. Why is AI used in 6G and what are its benefits?

AI (Artificial Intelligence) is used in 6G to automate network operations and improve performance. It can analyze large amounts of data and make smart decisions for network management. The benefits include automatic optimization, faster problem detection, reduced human effort, and improved efficiency. AI also supports advanced features like self-healing networks and zero-touch operations.

63. What is standalone 6G and why is it preferred?

Standalone 6G means the network works independently without relying on older technologies like 4G or 5G. It has its own complete infrastructure including RAN and Core Network. It is preferred because it reduces system complexity, improves performance, and provides better control over network operations. It also ensures faster deployment and a more unified system.

64. What is network slicing and why is it useful?

Network slicing is a technique where a **single physical network is divided into multiple virtual networks**, called slices. Each slice is designed for a specific service, such as healthcare, gaming, or IoT. It is useful because it allows different applications to get customized performance based on their needs, improving efficiency and user experience.

65. What is low latency and why is it important in 6G?

Low latency means very **small delay in data transmission**. In 6G, latency will be extremely low, which is important for real-time applications. For example, online gaming, virtual reality, autonomous vehicles, and remote surgery all require instant response. Low latency ensures smooth, fast, and reliable communication in these applications.

66. What are the main benefits of 6G technology?

6G offers many benefits such as ultra-high data speed, very low latency, better coverage, and high reliability. It also supports advanced technologies like AI, IoT, and immersive communication (AR/VR). Additionally, it focuses on energy efficiency and sustainability. These features will improve communication systems and enable new innovations in different industries.

67. What is automation in 6G and why is it needed?

Automation in 6G means the network can manage, configure, and optimize itself **using AI** and software tools. It is needed because **6G networks are very complex and cannot be managed manually**. Automation reduces human effort, increases efficiency, and ensures better performance. It also helps in achieving zero-touch network operations.

68. What is spectrum aggregation and why is it important?

Spectrum aggregation is the **process of combining multiple frequency bands to improve** network performance. In 6G, it helps achieve higher data rates and better coverage by using both low and high frequency bands together. It is important because it ensures efficient use of available spectrum and improves overall network

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69. What is 6G and why is it important?

6G is the sixth generation of mobile communication technology, which will succeed 5G. It is expected to provide extremely high-speed data transmission, ultra-low latency, and intelligent network capabilities. Unlike previous generations, 6G will not only focus on communication but also integrate computing, sensing, and AI-based services into a unified platform.

The importance of 6G lies in its ability to support future technologies such as smart cities, autonomous vehicles, remote healthcare, immersive communication (XR), and industrial automation. As society becomes more digitally dependent, 6G will play a key role in connecting people, devices, and systems efficiently. It will also help bridge the gap between the physical and digital worlds, making everyday life more connected, intelligent, and automated.

70. Why is security a major concern in 6G?

Security is a major concern in 6G because mobile networks are becoming a critical part of modern society. They are used in important sectors such as healthcare, transportation, industry, and government services. Any security failure in these systems can lead to serious consequences, including data breaches, financial loss, or even threats to human safety.

In 6G, the number of connected devices will increase significantly, including IoT devices, sensors, and automated systems. This expansion increases the attack surface for cyber threats. Additionally, new technologies like AI, cloud computing, and virtualization introduce new vulnerabilities. Therefore, strong security mechanisms are required to ensure data confidentiality, integrity, and availability, as well as to protect users' privacy.

71. Explain the concept of cyber-physical integration in 6G.

Cyber-physical integration refers to the merging of the digital (cyber) world with the physical world. In 6G, this means that physical systems such as machines, vehicles, sensors, and infrastructure will be closely connected with digital networks and software systems.

For example, smart factories may use sensors and AI systems to monitor and control machines in real time. Similarly, healthcare systems may use connected devices to track patient data remotely. This integration improves efficiency, automation, and decision-making.

However, it also introduces new security challenges because any cyber-attack can directly affect physical systems. Therefore, 6G must ensure strong security measures to protect both digital data and physical operations.

72. What role does Artificial Intelligence (AI) play in 6G security?

Artificial Intelligence (AI) plays a significant role in enhancing the security of 6G networks. AI can be used to automatically detect, analyze, and respond to cyber threats in real time. It helps in identifying abnormal behavior in the network, detecting malicious devices, and preventing attacks such as fake base stations or signaling attacks.

AI also improves network efficiency by enabling automation and self-learning capabilities. For example, AI can optimize network performance and predict potential failures before they occur.

However, the use of AI also introduces new challenges. AI systems must be secure, reliable, and resistant to attacks. If compromised, AI could make incorrect decisions or be manipulated by attackers. Therefore, ensuring trustworthy and explainable AI is essential for 6G security.

73. What is service availability and why is it important in 6G?

Service availability refers to the ability of the network to provide uninterrupted services to users under all conditions, including normal operations, system failures, or cyber-attacks. In 6G, many services will be critical for society, such as healthcare systems, emergency services, and smart infrastructure.

High availability ensures that users can rely on the network at all times. For example, in a smart healthcare system, a failure in network service could delay critical medical decisions.

To achieve this, 6G networks must be designed with resilience, redundancy, and strong protection against attacks like denial-of-service (DoS). Continuous monitoring and rapid response mechanisms are also necessary to maintain service reliability.

74. Explain network exposure and API security in 6G.

Network exposure in 6G refers to **allowing external applications and service providers to access network capabilities through APIs** (Application Programming Interfaces). This enables developers to create innovative applications by using network features such as data access, connectivity control, and performance optimization.

However, exposing network functions increases security risks. Unauthorized access or misuse of APIs can lead to data breaches or network disruption. Therefore, strong API security is essential.

API security includes mechanisms such as **authentication** (verifying identity), **authorization** (controlling access), and **monitoring** (tracking usage). In addition, **continuous threat analysis and runtime protection** are required to ensure safe API usage while maintaining user privacy.

75. How does 6G improve privacy and data protection?

6G introduces advanced mechanisms to enhance privacy and data protection compared to previous generations. It focuses on protecting data throughout its lifecycle, including data in transit (while being transmitted), data at rest (stored data), and data in processing.

One important concept in 6G is data ownership, where users have more control over their personal information. Secure identity management systems are used to ensure that only authorized users can access data.

Additionally, 6G is expected to adopt advanced encryption techniques, including quantum-resistant cryptography, to protect against future threats. Privacy-enhancing technologies and strict access control mechanisms will help ensure that sensitive information remains secure, even in complex and large-scale networks.

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76. What is 6G and what are its expected features?

6G is the sixth generation of mobile communication, expected to be deployed around 2030. It will go beyond 5G by integrating the digital and physical worlds. Key features include ultra-high data rates, extremely low latency, support for intelligent machines, and immersive technologies like holographic communication and extended reality (XR). It will also enable new services such as the Internet of Senses and advanced AI-driven networks.

77. Why is spectrum availability critical for 6G development?

Spectrum is essential because it determines the capacity, coverage, and performance of wireless networks. 6G requires significantly more spectrum due to increasing data traffic and new applications like holographic

communication and digital twins. Existing spectrum bands are insufficient, leading to a need for additional frequencies. Without timely spectrum allocation, the full potential of 6G cannot be achieved.

78. Explain the concept of the Internet of Senses.

The Internet of Senses is a 6G concept where digital communication extends beyond audio and visual data to include human senses such as touch, taste, and smell. It enables immersive experiences that closely replicate real-life interactions. This technology can reduce the need for physical travel, support remote healthcare, and improve virtual learning by creating realistic environments.

79. Describe holographic communication and its requirements. real-time 3D image communication

Holographic communication involves transmitting 3D images of people or objects in real time, creating a realistic interaction experience. It requires very high data rates (around 100 Mbit/s uplink and 1 Gbit/s downlink per user) and low latency to avoid motion sickness and ensure smooth visuals. It also demands large amounts of spectrum for wide-area coverage and mobility.

80. What is a massive digital twin and how does it impact spectrum needs?

A massive digital twin is a digital representation of physical environments such as cities, including buildings, traffic, and infrastructure. It relies on data from numerous sensors, which collectively generate high data traffic. Even though each sensor sends small data, the large number of sensors increases total demand, requiring significant spectrum, especially for uplink communication.

81. Explain the importance of the 7–15 GHz centimetric range in 6G.

The 7–15 GHz range is important because it offers a balance between coverage and capacity. It supports high data rates while still allowing wide-area mobility. This makes it suitable for key 6G use cases: mobile broadband, Internet of Senses, and digital twins. It is also closer to mid-band frequencies, allowing reuse of existing infrastructure and reducing deployment costs.

82. Discuss the role of the sub-THz range in 6G.

The sub-THz range (92–300 GHz) provides extremely large bandwidth, enabling data rates in the terabits per second (Tbps). However, it has limited coverage and mobility due to high signal attenuation. Therefore, it is mainly used for niche applications such as high-speed indoor communication, data centers, and device-to-device communication. It complements lower frequency bands rather than replacing them.

83. Differentiate between low-band, mid-band, and high-band spectrum.

Feature	Low-band Spectrum	Mid-band Spectrum	High-band (mmWave) Spectrum
Frequency range	Below 1 GHz	~1 GHz to 7 GHz	Above ~24 GHz
Coverage	Very wide (rural & deep indoor)	Moderate	Limited (short range)
Capacity/Data Rate	Low to moderate	Balanced	Very high
Main Use Cases	Rural connectivity, IoT, basic services	Mobile broadband, urban/suburban areas	Dense urban areas, hotspots, high-speed applications

84. What is the estimated spectrum requirement for 6G use cases?

For wide-area use cases like holographic communication and digital twins, approximately 3 GHz of spectrum is required. However, current and expected available spectrum falls short, with a gap of around 1.5–2.2 GHz. Additional spectrum, especially in the 7–15 GHz range, is needed to meet these demands.

85. Explain the pathway to achieving 6G spectrum.

The pathway involves collaboration between stakeholders such as governments, regulators, mobile operators, and researchers. Spectrum allocation decisions are made through global forums like the World Radiocommunication Conference (WRC). Standardization bodies like ITU and 3GPP define technical requirements. Timely planning and global harmonization of spectrum are essential to ensure successful deployment of 6G by 2030.